

Reviewer Pack — Hochster Regularity of Variable Co-Occurrence Graph Lower-Bo...

Entry e0a06bc1af46 · INCONCLUSIVE

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Hochster Regularity of Variable Co-Occurrence Graph Lower-Bounds Resolution Width

Entry ID: e0a06bc1af46 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-17 06:44:48 UTC

1. Conjecture

Field A (mathematical branch): Castelnuovo-Mumford regularity of squarefree monomial edge ideals via Hochster’s combinatorial formula on independence complexes (Hochster 1977; Bruns-Herzog 1993; Katzman 2006; Woodroffe 2014) — a corner of computational commutative algebra with essentially no proof-complexity-side publications (arXiv search ‘edge ideal regularity’ AND (‘resolution width’ OR ‘proof complexity’) returns no hits) **Field B** (complexity object): Resolution refutation width $w^*(F)$ for unsatisfiable 3-CNFs, viewed as a Ben-Sasson-Wigderson and Krajíček-style stepping stone toward size lower bounds for tree-like Resolution and Extended Frege (via the depth- $O(1)$ Frege simulation of Resolution)

Statement:

For an unsatisfiable 3-CNF F on $n \leq 10$ variables, build the variable co-occurrence graph $G(F)$ with $V = \{x_1, \dots, x_n\}$ and edges $\{x_i, x_j\}$ whenever x_i and x_j co-appear in at least one clause of F . Define $\text{reg}(F) := \max\{j \geq 0 : \exists W \subseteq V \text{ with reduced rational homology } \tilde{H}_{|W|-j-1}(\Delta\{G(F)[W]\}; \mathbb{Q}) \neq 0\}$, where $\Delta\{G[W]\}$ is the independence complex of the induced subgraph $G[W]$ (Hochster’s formula for $\text{reg}(Q[x]/I_{\{G(F)\}})$). Conjecture: for every unsatisfiable 3-CNF F , the minimum Resolution refutation width satisfies $w(F) \geq \text{reg}(F) + 2$; equivalently, a single F with $w(F) < \text{reg}(F) + 2$ falsifies it.

Rationale (proposer’s reasoning):

Hochster’s formula recasts an algebraic invariant of the edge ideal as the maximum homological complexity over induced subgraphs, identifying variable clusters whose mutual logical entanglement is topologically nontrivial; intuitively, no narrow refutation can simultaneously break such a cluster, forcing high resolution width. The bridge from edge-ideal regularity to proof complexity is essentially unstudied, and unlike single-number truth-table invariants (calibration trap (j)), $\text{reg}(F)$ is a *relational* invariant of the CNF graph computed by quantifying over induced substructures, sidestepping the Razborov-Rudich largeness criterion that defeats truth-table approaches. If valid, BSW gives size $\geq \exp(\Omega(\text{reg}(F)^2/n))$, and via Krajíček’s depth- d Frege simulation this propagates to non-trivial EF line-count certificates.

Taxonomy category: PROOF_COMPLEXITY_EXT_FREGE (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): deac4029347f65c7

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

Across 90 unsat 3-CNFs ($n \in \{6, 8, 10\}$, 30 seeds each, $\alpha=4.5$), compute $\text{reg}(F)$ via Hochster on $G(F)$ and $w(F)$ via *width-bounded resolution saturation*. *Conjecture SUPPORTED iff every seed satisfies $w(F) \geq \text{reg}(F)+2$; FALSIFIED by any single seed with $w^*(F) < \text{reg}(F)+2$.*

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.92	SAFE	SAFE
NATURAL_PROOFS	SAFE	0.85	SAFE	SAFE
ALGEBRIZATION	SAFE	0.90	SAFE	SAFE
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - edge ideal regularity independence complex resolution width proof complexity - Hochster formula Castelnuovo-Mumford regularity CNF refutation lower bound - co-occurrence graph monomial ideal Betti numbers resolution width unsatisfiable

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import sys
import random
import math
import itertools
from collections import defaultdict

def generate_3cnf(n, m, seed):
    random.seed(seed)
    variables = list(range(1, n + 1))
    clauses = []
    for _ in range(m):
        clause = random.sample(variables, 3)
        clause = [(var, random.choice([True, False])) for var in clause]
        clauses.append(clause)
    return clauses

def is_unsatisfiable(F, n):
    assignments = itertools.product([False, True], repeat=n)
```

```

for assignment in assignments:
    satisfied = True
    for clause in F:
        clause_satisfied = False
        for var, sign in clause:
            if assignment[var - 1] == sign:
                clause_satisfied = True
                break
        if not clause_satisfied:
            satisfied = False
            break
    if satisfied:
        return False
return True

def build_co_occurrence_graph(F, n):
    graph = defaultdict(set)
    for clause in F:
        variables = [var for var, _ in clause]
        for i in range(len(variables)):
            for j in range(i + 1, len(variables)):
                graph[variables[i]].add(variables[j])
                graph[variables[j]].add(variables[i])
    return graph

def find_independent_sets(graph, W):
    if not W:
        return [frozenset()]
    v = W[0]
    independent_sets = []
    for subset in find_independent_sets(graph, W[1:]):
        independent_sets.append(subset)
        if all(u not in subset for u in graph[v]):
            independent_sets.append(subset | {v})
    return independent_sets

def compute_betti_numbers(independent_sets, k):
    if k < 0 or k >= len(independent_sets):
        return 0
    boundary_matrix = [[0] * len(independent_sets) for _ in range(len(independent_sets))]
    for i, sigma in enumerate(independent_sets):
        for j, tau in enumerate(independent_sets):
            if sigma.issuperset(tau) and len(sigma) == len(tau) + 1:
                boundary_matrix[i][j] = 1
    rank = 0
    for col in range(len(boundary_matrix[0])):
        pivot = -1
        for row in range(rank, len(boundary_matrix)):
            if boundary_matrix[row][col] == 1:
                pivot = row
                break
        if pivot == -1:
            continue
        boundary_matrix[rank], boundary_matrix[pivot] = boundary_matrix[pivot], boundary_matrix[rank]
        for row in range(len(boundary_matrix)):
            if row != rank and boundary_matrix[row][col] == 1:

```

```

        for c in range(len(boundary_matrix[0])):
            boundary_matrix[row][c] ^= boundary_matrix[rank][c]
        rank += 1
    return len(boundary_matrix) - rank

def compute_reg(F, n):
    graph = build_co_occurrence_graph(F, n)
    max_reg = 0
    for W in itertools.chain.from_iterable(itertools.combinations(range(1, n + 1), r) for r in range(1, n + 1)):
        independent_sets = find_independent_sets(graph, list(W))
        for j in range(len(W) + 1):
            if compute_betti_numbers(independent_sets, len(W) - j - 1) > 0:
                max_reg = max(max_reg, j)
    return max_reg

def resolution_width(F, n):
    clauses = [frozenset((var, sign) for var, sign in clause) for clause in F]
    closure = set(clauses)
    w = 2
    while True:
        new_clauses = set()
        for c1, c2 in itertools.combinations(closure, 2):
            common_vars = set(var for var, _ in c1) & set(var for var, _ in c2)
            for var in common_vars:
                if any((var, True) in c1 and (var, False) in c2) or any((var, False) in c1 and (var, True) in c2):
                    new_clause = (c1 | c2) - {(var, True), (var, False)}
                    if len(new_clause) <= w:
                        new_clauses.add(new_clause)
            if not new_clauses:
                break
        closure.update(new_clauses)
        if any(len(closure) == 0 for clause in closure):
            return w
        w += 1
    return float('inf')

def run_trial(seed):
    n = random.choice([6, 8, 10])
    m = int(4.5 * n)
    F = generate_3cnf(n, m, seed)
    if not is_unsatisfiable(F, n):
        return {
            "metric_name": "w*(F) - reg(F)",
            "metric_value": 0,
            "instances_tested": 1,
            "conjecture_holds": True,
            "counterexample": ""
        }
    reg = compute_reg(F, n)
    w = resolution_width(F, n)
    metric_value = w - reg
    conjecture_holds = metric_value >= 2
    counterexample = "" if conjecture_holds else f"w*(F) = {w} < reg(F) + 2 = {reg + 2}"
    return {
        "metric_name": "w*(F) - reg(F)",
        "metric_value": metric_value,

```

```

        "instances_tested": 1,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] if len(sys.argv) > 1 else [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97]
    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    metric_values = [r["metric_value"] for r in results]
    mean = sum(metric_values) / len(metric_values)
    std = math.sqrt(sum((x - mean) ** 2 for x in metric_values) / len(metric_values))
    support_fraction = sum(r["conjecture_holds"] for r in results) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean} std={std} support_fraction={support_fraction}")
    else:
        for r in results:
            if not r["conjecture_holds"]:
                print(f"RESULT: FALSIFIED counterexample={r['counterexample']} first_failing_seed={seed}")
                break

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

TRIAL: {'metric_name': 'w*(F) - reg(F)', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': True}
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d752578.py", line 149, in <module>
    result = run_trial(seed)
              ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d752578.py", line 133, in run_trial
    w = resolution_width(F, n)
        ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d752578.py", line 108, in resolution_width
    if any((var, True) in c1 and (var, False) in c2) or any((var, False) in c1 and (var, True) in c2):
    ~~~~~^
TypeError: 'bool' object is not iterable

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed with a `TypeError` in `resolution_width` (misuse of `any()` on a boolean expression), producing only 1 trial instead of the pre-registered 90 seeds. No valid data was collected to evaluate the support condition. | next: Fix the `resolution_width` function (the `any()` calls wrap non-iterable boolean expressions) and rerun the full 90-seed sweep across $n \in \{6, 8, 10\}$.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	claude_max	opus	0	0	233747
2	preregistration	claude_max	opus	0	0	4643
3	novelty	claude_max	opus	0	0	3112
4	novelty	claude_max	opus	0	0	5608
5	test_gen	mistral	codestral-latest	0	0	325983
6	test_gen	mistral	codestral-latest	0	0	311288
7	test_gen	mistral	codestral-latest	0	0	311928
8	test_gen	mistral	codestral-latest	0	0	313959
9	judge	claude_max	opus	0	0	4272

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 1514541 ms total latency. Provider mix: {'claude_max': 5, 'mistral': 4}

(full prompt+response transcripts available in `research/audit/e0a06bc1af46.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/e0a06bc1af46.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/e0a06bc1af46.tar.gz` (if generated)